

INTEGRATING MOBILE EDGE COMPUTING INTO UNMANNED AERIAL VEHICLE NETWORKS: AN SDN-ENABLED ARCHITECTURE

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ABSTRACT

Over the past few years, we have witnessed an exponential increase in demand for seamless and continuous Internet access to online services — multimedia services, online business, gaming, scientific analysis, and so on. With the significant advances in computing and communication techniques of edge devices, the mobile edge computing (MEC) paradigm has emerged and has been demonstrated to be more feasible and efficient in providing wireless Internet access or computing in some scenarios compared to the traditional cloud computing platform. Furthermore, the promotion and popularization of unmanned aerial vehicles (UAVs) in recent years provides the promise of ubiquitous data sensing, leading to an era of aerial MEC. In this article, we first present four representative architectures for the UAV-based MEC system. Then we survey some recent research products in this area. In particular, we argue that the UAV can be regarded as a MEC unit and propose the paradigm of a multi-UAV-based MEC network. Furthermore, we employ software-defined networking to improve the network's scalability and controllability and present the framework of the software-defined UAV network. Based on the proposed framework, we introduce the concept of a "smart function table" that can be applied to execute the network operation in the proposed framework. After evaluating our proposed framework, especially concerning the cooperative flying ability and control delay, several challenging issues and future research suggestions are presented to guide this direction.

INTRODUCTION

The revolution of the Internet of Things (IoT) technology provides ubiquitous Internet service access for mobile users (MUs) and promotes numerous civil/military applications, ranging from deep sea to outer space [1]. As a typical example, IoT devices collect data from the physical world and deliver it to the remote cloud server for further processing via heterogeneous wired or wireless networks. Recently, successful developments in information and communications technology (ICT) have brought about vast amounts of services sensitive to delay, which challenges both the computing platform and the data transfer architecture [2]. For instance, current remote cloud computing platforms are usually deployed at the core of the Internet, which is not adaptive to delay-sensitive services since the propagation delay between the end devices and the cloud server is huge.

As an alternative option, the mobile edge computing (MEC) paradigm is more suitable for delay-sensitive services due to the following advantages:

1. The MEC unit is usually deployed at the edge of the network, which is closer to the end device than the cloud center.
2. The MEC technique can provide timely data computing, which can satisfy the quality of experience (QoE) of most lightweight data-driven services.
3. Meanwhile, the MEC technique can mitigate the traffic pressure in the Internet backbone [3, 4].

With the ongoing expansion of automation, robot technology, and so on, unmanned aerial vehicle (UAV)-aided MEC systems have been introduced and gained growing attention from academia and industry. UAV-aided MEC systems integrate UAV into the MEC system, where the UAV can work in

multiple modes according to real application scenarios [5]. For instance, in some scenarios, the UAV can be a communication relay between the terrestrial MEC unit and the MUs. In contrast, the UAV can sometimes be treated as an aerial MEC unit that can offload computation tasks from the terrestrial MEC unit. Compared to the normal terrestrial MEC systems, the UAV-based MEC system produces higher adaptability and can be widely applied in the barren areas (desert, rainforest, etc.), battlefields, disaster sites, hotspot areas, and so on. In contrast, the terrestrial MEC system is hard to deploy and easy to destroy [6]. It can be speculated that the UAV-based MEC system will play an essential role in the future space-terrestrial information network or end-edge-cloud system. Furthermore, to improve the scalability and functionality of the UAV-based MEC system, some works are focusing on multi-UAV-based MEC systems or multi-UAV-enabled MEC networks where each UAV is a distributed MEC server with a distinguished network function and can cooperate to perform a joint network function (e.g., data sharing/routing or data distributed computing among the UAVs) [7].

However, even if the multi-UAV-based MEC network can provide more comprehensive and complex MEC services, the incidental challenging issues have to be confronted and tackled:

1. The differentiation of the UAV or the complexity of the multi-UAV-based MEC network enables no uniform network management platform.
2. The multi-UAV-based MEC network is a mobile ad hoc network whose network topology varies over time, resulting in the network function not being easy to control cooperatively [8, 9].

Recently, the idea of "software is eating the world" promotes network function virtualization, software defined networking (SDN), and others, which are considered as potential solutions to overcome the aforementioned challenges [6]. SDN separates the network control platform from the network function component (e.g., the switch in the computer network). In SDN, the network function devices are only responsible for executing the network functions (e.g., data forwarding), and all the network policies are determined and programmed at the uniform network control platform [10]. This enables centralized network control and allows the deployment of complex and cooperative network policies.

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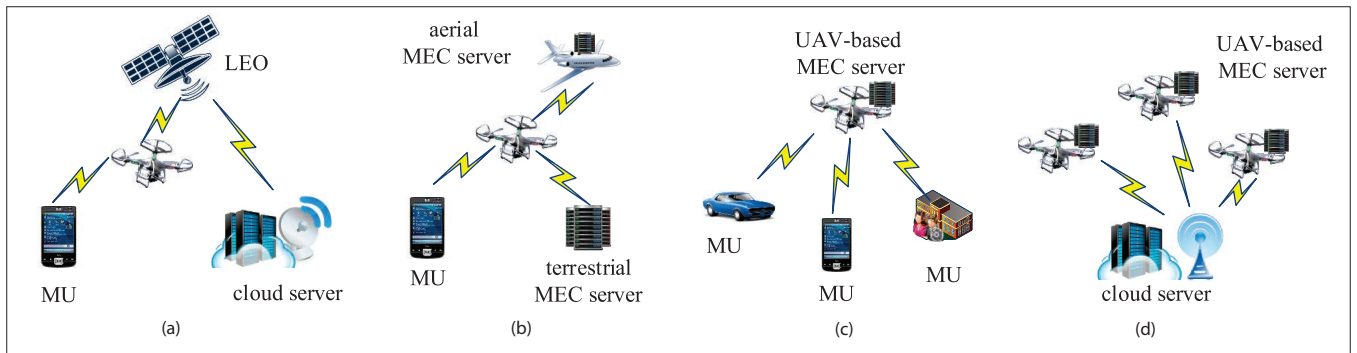


FIGURE 1. Four architectures for UAV-based MEC system: a) relay between MU and cloud server; b) relay between MU and terrestrial/aerial MEC server; c) MEC server; d) distributed MEC in the end-edge-cloud architecture.

In this work, four representative architectures for the UAV-based MEC system are presented. After that, a comprehensive survey is given on the state-of-the-art research products about the aspect of the UAV-based MEC system [11]. In particular, in this article, we consider the multi-UAV-based MEC network consisting of heterogeneous UAV-based MEC units. Motivated by the SDN technique, we utilize SDN to improve the scalability and programmability of the MEC network and propose a framework for integrating MEC into the software-defined UAV network. Furthermore, we highlight some challenges and open research directions about this topic.

The remainder of this article is organized as follows. We present the major existing solutions on this topic. We present our proposed framework, which integrates MEC into the software-defined UAV network. We present some selected evaluation results. We discuss possible future directions and open issues about this concept, followed by concluding remarks.

OVERVIEW OF MAJOR EXISTING SOLUTIONS

In this section, we introduce some recent advances and research products/efforts in the UAV-based MEC system.

As presented in Fig. 1, there usually are four categories of architectures for UAV-based MEC systems. The UAV has multiple roles ranging from the communication/computation relay to the MEC server. The original scenarios are shown in Fig. 1a, in which the UAV is the communication relay between MUs and cloud servers. This scenario is common and widely deployed at present, but it is limited by the long propagation delay between MUs and remote cloud servers. With the emergence of MEC, the UAV works as the relay between MUs and the terrestrial/aerial MEC server; that is, the UAV is only in charge of transferring the service request to the closer MEC server, as shown in Fig. 1b. In recent years, with the successful development of computing and communication units, the UAV-based MEC unit has emerged, as shown in Fig. 1c: the UAV itself can deal with lightweight computation tasks. This also promotes many complex paradigms, such as the computation task distribution among the UAVs (e.g., the scenario in Fig. 1d).

The aforementioned four architectures are designed for different application scenarios with other purposes. The first one (Fig. 1a) is a robust and stable architecture that has been widely applied in environmental monitoring and military surveillance. The second one (Fig. 1b) is a transitional product where the communication range of the terrestrial/aerial MEC server is limited, and the UAV is equipped with computing/storing components of lower ability. The third architecture (Fig. 1c) is suitable for cases where the service requests for lightweight data computation are frequent, and the requested services are susceptible to delay. The fourth case (Fig. 1d) is dedicated to the scenario where the UAV flock or multi-UAV-based MEC server is introduced to perform distributed task computation and deployment. This architecture is more intelligent and adaptive to complex scenarios (e.g., heavyweight task computation).

Some recent research products related to UAV-based MEC systems are presented as the following.

In [8], the authors introduce the UAV into the MEC network and treat the UAV as a relay mobile MEC node to address the issue of constricted computing resources in the traditional MEC network. To optimize the computing energy, the authors propose a three-layer computation offloading strategy by predicting the mission location. The UAV can be scheduled to the optimal position to perform mission deployment. In [12], the authors argue that the UAV can be regarded as a MEC server in a forest fire monitoring system. The authors propose a framework integrating the MEC server, that is, the UAV and Industrial Internet of Things (IIoTs) sensors that are dedicated to collecting the data. Meanwhile, the authors propose a learning method based on Markov random field (MRF) and Swarm Optimization Algorithm (SOA) to allocate computing resources, especially when the data priority is considered. In [9], the authors propose a framework for a MEC network where the UAV is scheduled and controlled by cellular ground base stations (GBSs) to perform computation offloading. To minimize the UAV's mission achievement time, the authors propose an optimal algorithm to when the speed constraint of the UAV, the UAV's trajectory together with the data scale of the mission are concurrently considered, such that a high-quality sub-optimal solution can be acquired. Different from the aforementioned proposals that offload the mission from the MEC network to the UAV, in [13], the authors study the data eavesdropping issues when the mission is migrated from the UAV to the MEC network. The authors build a set of mathematical formulations and transform the problem into a series of convex issues that can guarantee active and proactive eavesdropping. In [10], the authors propose an air-ground network architecture integrating 5G, MEC components, and UAV where the UAV is treated as a service access platform. In particular, the authors propose an intelligent charging-offloading scheme to minimize the service providing delay. To improve the running efficiency of the proposed scheme, the authors utilize the block-coordinate descent (BCD) method to decompose the proposed scheme into a series of convex sub-problems so that an improved greedy algorithm can be achieved.

In [14], the authors propose the use of a UAV to migrate task computation from the terminal devices to the wireless access points (i.e., the UAVs), especially for delay-sensitive tasks. Different from the proposals in [8, 9], the authors consider both the communication and computation consumed energy on the UAVs and propose a Lagrangian duality method and SCA-based algorithm to minimize the total energy. Recently, there are also some multi-UAV-based proposals that embed the UAV flock into the MEC network. For instance, in [11], the authors treat the UAVs as MEC nodes and propose the concept of distributing computing tasks among the UAVs. To balance the computing load among the UAVs, the authors propose a multi-UAV deployment mechanism based on differential evolution and a deep reinforcement learning (DRL)-based task scheduling scheme.

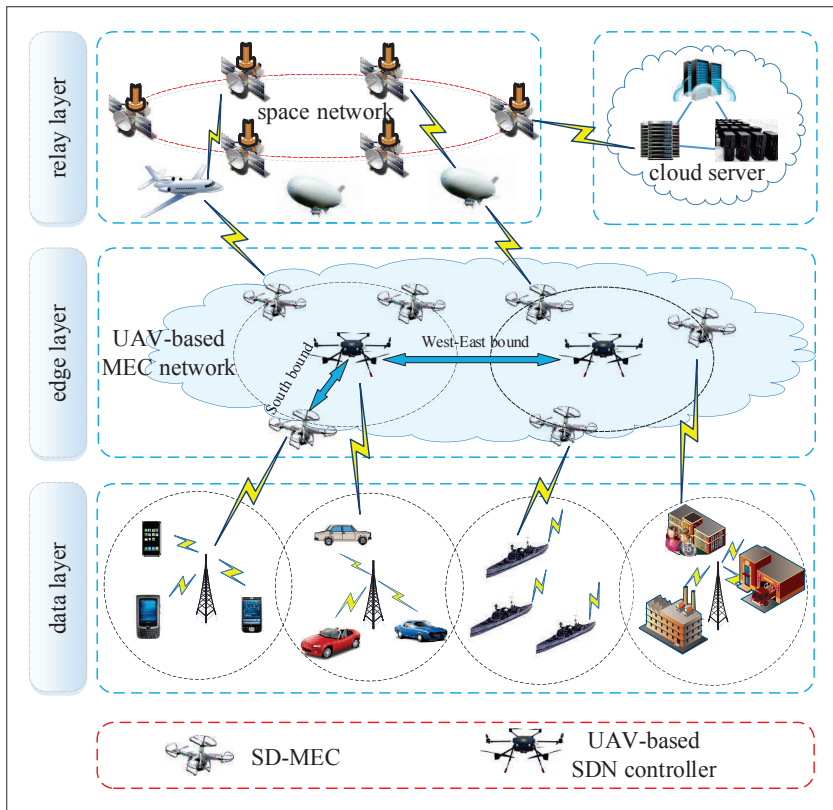


FIGURE 2. Proposed software-defined UAV-based MEC network.

As mentioned above, the multi-UAV-based MEC architecture, benefitting from its high performance and scalability, is more adaptive to heavyweight data services. Recently, to provide diversified and complex MEC services, some researchers have employed and proposed the concept of integrating each UAV into a united network. Thus, the UAVs can communicate with each other by following pre-defined communication protocols and perform cooperative network functions. However, even if it can provide the ability of cooperative network operation, it brings additional burdens to both the system control and data transfer components, which results in the following two challenging issues:

1. The multi-UAV-based MEC network is a mobile ad hoc network with dynamically changing network topology which is hard to monitor.
2. The multi-UAV-based MEC network based on traditional network architecture has a distributed control plane.

This cannot provide an abstract view of the entire system, enabling no deployment of cooperative and exact network functions.

SOFTWARE-DEFINED UAV-BASED MEC NETWORK

In this article, we take the architectures in both Figs. 1c and 1d into account and introduce the concept of multi-UAV-based MEC networks. To perform cooperative network functions for the multi-UAV-based MEC network, we employ SDN to optimize the architecture of the multi-UAV-based MEC network. With SDN features, the network operation plane can be decoupled from the network control plane so intelligent and scalable MEC service access can be provided. In this section, the UAV-based networking system's proposed architecture is first presented, followed by the state for a proposed smart operation/flow table framework.

PROPOSED NETWORK ARCHITECTURE

This section proposes an SDN-enabled architecture that can support centralized management for the multi-UAV-based MEC network integrating multiple UAVs or a UAV flock. By SDN, the

multi-UAV-based MEC network can be divided into three functional layers, as shown in Fig. 2.

Data Layer: The data layer of the proposed network architecture is dedicated to collecting data from the ubiquitous communication networks (e.g., WiFi networks for hotspots, ad hoc networks, and vehicular networks). In particular, we assume the data layer of the proposed network architecture can determine the QoS or QoE (i.e., the priority or the delay sensitivity of the sensed data) and can provide exact localization for MUs within it. Furthermore, the MUs in the data layer are restricted from communicating and can access the edge layer of the proposed network architecture in real time.

Edge Layer: As shown in Fig. 2, different from the current research advances (e.g., the proposal in [8, 9]), the edge layer of the proposed network architecture can be regarded as a network integrating UAV-based MEC units of two normal types: the software-defined MEC server (SD-MEC) and the UAV-based SDN controller. The edge layer is actually a distributed two-layer network where the SD-MEC works in the first layer, and the SDN controller is in the second layer. Different from the proposal in [8, 9], in our work, the SD-MECs can communicate with each other via dedicated communication channels and can be allocated to different local area networks according to their features, functions, or positions. For instance, in Fig. 2, the UAVs can be divided into two local networks. The deployment range of each local network can be determined by referring to the access frequency of the MUs in each local area of the data layer. Each local network is under the management of one SDN controller via the pre-defined southbound protocol. Note that in the proposed network architecture, the SD-MEC provides direct MEC access to the MUs in the data layer. At the same time, the SDN controller cannot directly get the service request from the MUs and is only responsible for controlling the SD-MEC in each local UAV-based MEC network such that each SDN controller can keep a local view of the local UAV-based MEC network, even for the MUs in the data layer.

Furthermore, apart from the two-layer-based physical architecture of the edge layer, the edge layer of the proposed network architecture can also provide the two-layer-based functional request mode. The procedure is the following: A vehicle in the data layer requests MEC service, for example, automatic driving policy computation from a handy SD-MEC in the edge layer. If the requested SD-MEC can determine the automatic driving policy, the SD-MEC directly deploys the policy at the MU. The requested SD-MEC will request the SDN controller to schedule the network resource to determine the automatic driving policy. For instance, the SDN controller synchronizes the information from each SD-MEC within its local network, determines the policies by itself, or deploys distributed computation policies at each corresponding SD-MEC, as in the architecture shown in Fig. 1d. Furthermore, the SD-MECs in multiple local networks can be jointly scheduled to execute one joint mission for the services across various areas. Thus, it requires data synchronization among the SDN controllers via the dedicated west/eastbound protocol.

Relay Layer: The relay layer, together with the edge layer, actually constitutes the end-edge-cloud architecture. The relay layer of the proposed network architecture aims to provide the data transfer relay for services that are not very sensitive to delay or services whose QoE cannot be met by the edge layer's computing ability. As shown in Fig. 2, the SD-MECs in the edge layer of the proposed network architecture deliver the service requests together with the corresponding data to the remote cloud server via the low Earth orbit (LEO) network or

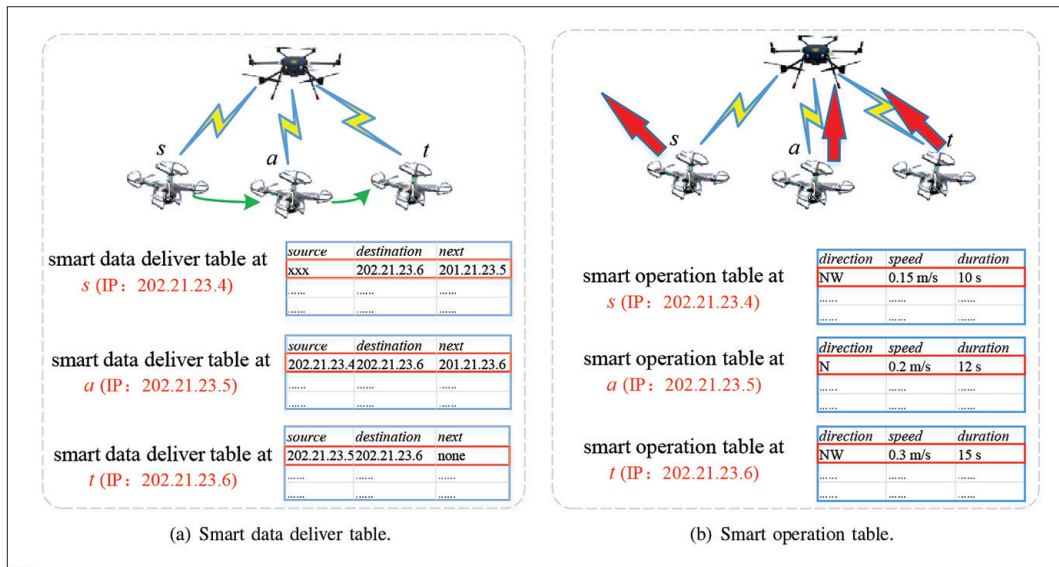


FIGURE 3. Smart function table: a) smart data delivery table; b) smart operation table.

the relay airship in the stratosphere. Thus, on account of the high-performance computing ability, the remote cloud server can successfully compute and determine the policies to provide the services, leading to the end-edge-cloud paradigm.

SMART FUNCTION TABLE

As the most typical paradigm, the OpenFlow protocol is famous for its applicability and scalability. It has been widely promoted in many local area networks (e.g., the Google B4 network). OpenFlow deploys and executes the network functions (i.e., the data transfer function) by deploying the so-called flow table. For instance, to deploy the data routing policy, the action field of the flow table specifies the data output port at each corresponding port when the corresponding data traverses each switch. In this article, to achieve exact and cooperative functions for the multi-UAV-based MEC network, the concept of a smart function table is introduced by referring to the flow table mechanism of OpenFlow. As shown in Fig. 3, the smart function table includes two types: the smart data delivery table and the smart operation table. Like the flow table of OpenFlow, the smart function table is derived from the corresponding services determined and provided by the SDN controller.

Smart Data Deliver Table: The smart data delivery table is dedicated to providing data transfer or synchronization among UAVs, which is similar to the flow table concept in OpenFlow. This is adaptive to scenarios where the UAV-based MEC units have discrepant network functions. For instance, in Fig. 3a, apart from providing MEC services, UAV t is also equipped with a high-performance data storing component. Thus, to provide scalable network functions, UAV t only functions as data storage, while the other normal SD-UAVs in the local network work as the service provider and can support the ability of data routing. As shown in Fig. 3a, UAVs s (IP: 202.21.23.4 and IP: 202.21.23.5) are the normal SD-MEC, and t (IP: 202.21.23.6) is the data storage. The tables shown in Fig. 3a are the corresponding smart data delivery tables for performing data transfer from UAVs s to UAV t .

Smart Operation Table: Unlike the smart data delivery table, the smart operation table aims to control or adjust the flying features of each UAV in each UAV-based local network to provide stable and seamless MEC services. For instance, in Fig. 3b, under the control of the SDN controller, UAVs s , a , and t are uniformly scheduled to move along a pre-defined trajectory (corresponding to a service providing policy) to reach the target position for providing the MEC services. In Fig. 3b, UAVs s , a , and t are specified with moving directions of northwest

(NW), north (N), and NW, respectively. The speed of UAVs s , a , and t are 0.15 m/s (with duration 10 s), 0.2 m/s (with duration 12 s), and 0.3 m/s (with duration 15 s), respectively.

EVALUATION

In this section, we present some evaluation results to demonstrate the availability of the proposed scheme. Due to limited space, we only test the cooperative flying performance of the UAV flock when the smart operation table of the proposed framework is executed. The simulations were conducted on an Intel® Core i7-10510U 1.80 GHz machine with 40 GM memory within 200 time slots. We use Matlab R2018a to simulate the framework in a 3D area of $150 \text{ m} \times 120 \text{ m} \times 120 \text{ m}$ with potential air traffic control areas. Ten SD-MECs are uniformly controlled under one common UAV-based SDN controller. Note that the current communication ability of a UAV can provide HD image transmission of at most 7 km. Hence, the exact cooperative operation can be conducted in the simulation scenario. The flock control model follows Boid's model based on artificial potential field theory. In Fig. 4, the red star denotes the UAV-based SDN controller, the hollow blue circle is the SD-MEC, and the colorful solid circles are the air traffic control areas. In particular, the dotted line presents the SD-MEC under the control of the SDN controller, and the line with the arrow presents the flying direction of the SDN controller. The cruising relationship between SDN controller and SD-MEC follows the Gravitation-Repulsion model of the artificial potential field. The SDN controller adjusts the flying attitude of each SD-MEC by localizing each SD-MEC and deploying the smart operation table. The evaluation procedure is divided into eight phases; the results are shown in Fig. 4. Our proposed framework is available to efficiently schedule each SD-MEC's flying altitude to provide seamless and larger-range MEC access. Meanwhile, due to the nature of SDN, it can be speculated that the proposed smart data delivery table can achieve exact data transfer or sharing among the SD-MECs such that distributed edge computing services can be provided.

Furthermore, we also test the control delay together with the delay variance it takes to perform the cooperative flying conducted in Fig. 4. We compare our proposed scheme with the distributed control approaches as presented in [15]. Note that the state of each SD-MEC is periodically synchronized every five time slots. The control delay is defined as the maximum data propagation delay between the UAV-based SDN controller and the SD-MECs or among the SD-MECs (in the distributed approach). The control delay definition derives from the fact that the accuracy of cooperative network control depends on

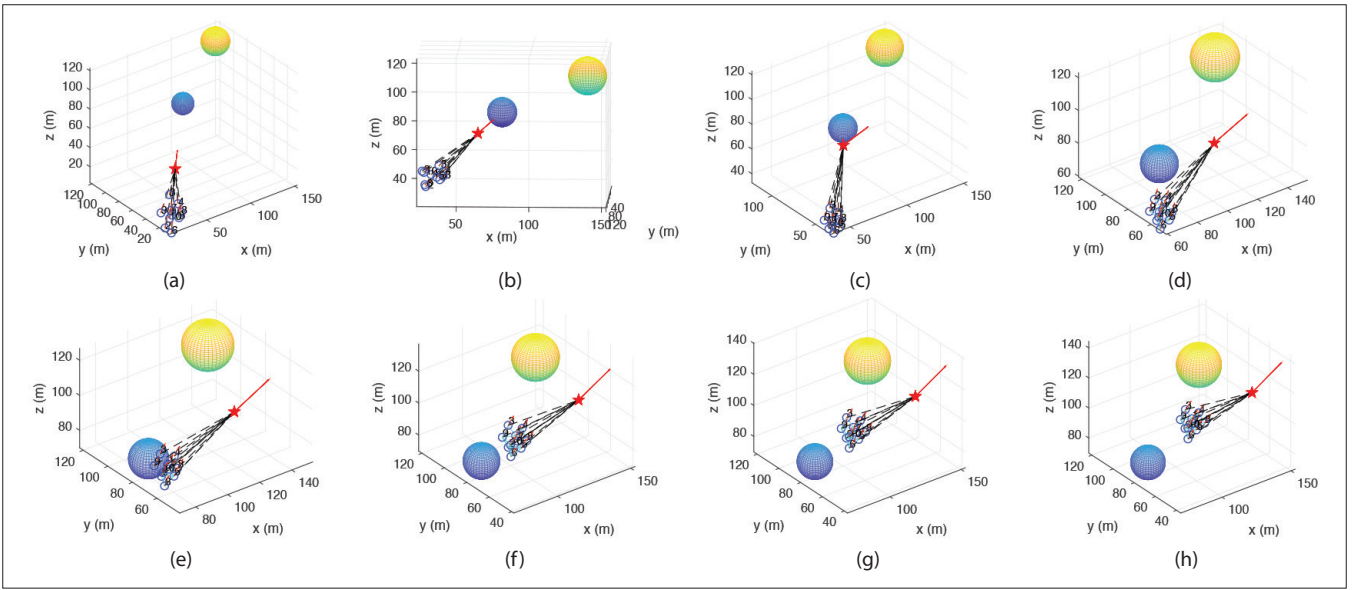


FIGURE 4. Test for the cooperative flying ability based on the proposed smart operation table: a) Phase 1; b) Phase 2; c) Phase 3; d) Phase 4; e) Phase 5; f) Phase 6; g) Phase 7; h) Phase 8.

all the synchronous information from each SD-MEC (i.e., the largest data propagation delay among the SD-MECs).

As the results in Figs. 5a and 5b show, our proposed scheme requires larger control delay to synchronize the information. It can be inferred from Fig. 4 that the factors are that the UAV-based SDN controller is not deployed at the feasible place and is far from the SD-MEC flock to highlight the key idea of our proposed scheme. Thus, the control delay can be decreased when the UAV-based SDN controller is adjusted to the right place, which is considered as a potential future task related to the topic of this work. In comparison, the delay variance the proposed scheme takes is far less than the distributed approach. This is due to the UAVs not being uniformly scheduled. Hence, it can be inferred that our proposed scheme is more suitable for multi-UAV cooperation.

POTENTIAL FUTURE DIRECTIONS AND CHALLENGES

The software-defined UAV-based MEC network is a promising paradigm for providing ubiquitous Internet access and improving the QoE of MUs. The following are some challenges and possible future directions on this topic.

SOUTHBOUND INTERFACE

The southbound interface is a protocol between the network data plane and control plane (corresponding to the SD-MEC and the UAV-based SDN controller). Unlike OpenFlow, OVSDB, and PCEP in traditional computer networks, the first challenging issue is standardizing the interaction between the SD-MEC and the UAV-based SDN controller (via dedicated wireless communication mode). Different from the OpenFlow-based computer networks via the dedicated wired control tunnel, the interaction between the SD-MEC and the UAV-based SDN controller has to be re-designed. This is because the software-defined UAV-based MEC network is a mobile ad hoc network where the control tunnel via a wireless control tunnel is not stable. Hence, it is indispensable to design a dedicated interaction method (including function request from the SD-MEC, data synchronization between the SD-MEC and the UAV-based SDN controller) to perform network functions. Furthermore, the southbound interface should also collect the network states or information from the SD-MECs and deploy the network function policies to the SD-MECs. Furthermore, following the concept earlier, the southbound interface should also support the ability to provide distinguished network functions, including data delivery function and UAV trajectory cooperative scheduling function.

RESOURCE ALLOCATION

Resource allocation is the most challenging issue in the software-defined UAV-based MEC network. It refers to the concepts of resource allocation in the SDN controller, resource allocation among SD-MECs, and resource allocation between the SD-MEC and cloud server, respectively.

Resource Allocation in the SDN Controller: In a software-defined UAV-based MEC network, the SD-MECs have to frequently request the SDN controller for network function policies. As a reply, the SDN controller has to frequently reconfigure the smart operation table on each SD-MEC. This results in additional waste of both the communication and computation resources. Hence, it is a nontrivial task to optimize the resource scheduling in the SDN controller.

Resource Allocation among SD-MECs: In a software-defined UAV-based MEC network, the resources on each SD-MEC should be efficiently allocated, mainly when multiple SD-MECs cooperate in executing a network function or the QoE of numerous network services is concurrently taken into account.

Resource Allocation between SD-MEC and Cloud Server: When the end-edge-cloud system has to be adopted to provide cooperative MEC service, it is indispensable to optimize the resource allocation at each component according to each embedded device's features. For instance, it is more feasible for SD-MEC to provide services sensitive to delay but hard to access by the service provider. At the same time, the remote cloud server can be scheduled to execute a large-scale mission that has no severe requirement on delay.

COLLABORATION MECHANISM

Unlike traditional single-UAV-based MEC systems, the collaboration issues in software-defined UAV-based MEC networks cannot be ignored. Apart from the collaboration between the SD-MECs and terrestrial MEC server as most previous studies discussed, the collaboration issues in the UAV-based MEC network includes two aspects.

Collaboration between SD-MEC and SDN Controller: The edge layer of the proposed software-defined UAV-based MEC network is a network with a dynamic network topology. This implies that the control messages from the SDN controller cannot always be concurrently received by SD-MEC, resulting in the network function not being able to be cooperatively deployed. Hence, it is a challenging issue to design the collaboration mechanism to guarantee network cooperative control policies.

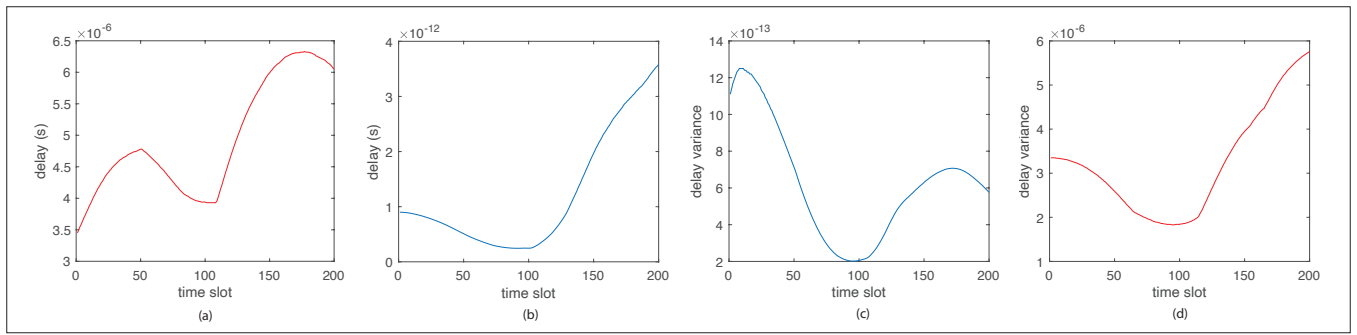


FIGURE 5. Test for the control delay and the variance for the control delay: a) control delay test for the proposed scheme; b) control delay test for the distributed approach in [15]; c) delay variance test for the proposed scheme; d) delay variance test for the distributed approach in [15].

Collaboration among SD-MECs: The SD-MECs or the UAV-based local network is a UAV flock. To efficiently schedule the UAV flock for providing seamless MEC service, the position of each UAV has to be frequently adjusted. Hence, it is indispensable to design a dedicated collaboration mechanism to control the flight attitude of the UAV flock.

CONCLUSIONS

Compared to terrestrial MEC systems, the UAV-based, especially multi-UAV-based, MEC system, is expected to greatly improve Internet services' QoE. In this article, we first discuss the typical application scenarios for the UAV-based MEC system. To acknowledge the advantages and limitations of this domain, we summarize and discuss four typical architectures for the UAV-based MEC system. We give a comprehensive survey about current research advances in this area for the four typical architectures. Unlike most of the current proposals, we regard the multi-UAV-based system as a network and utilize the SDN paradigm to improve it. We propose a smart function table framework to perform cooperative network functions, including the smart data delivery table and the smart operation table. Evaluation results demonstrate that our proposed framework can cooperatively control UAVs to provide ubiquitous MEC services. Finally, possible future directions and challenging issues are provided for the enlightenment of future researchers in this area.

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BIOGRAPHIES

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